Unimodality for one occurrence of the pattern 112

MathDB 345816, Task 1

Principia Math In Progress

22 September 2026

Dorward's Conjecture 4.1 [1] follows by identifying its generating polynomial with that of a parabolic quotient of the Weyl group of type B. The required unimodality theorem is Stanley's classical Theorem 3.1 [2].

Theorem 1. *For every $n \geq 3$, the polynomial $\mathrm{LS}_{n,1}(112)$ is unimodal. More precisely, put $m = n-2$ and $[r]_q = 1 + q + \cdots + q^{r-1}$. Then*

$$q^{-1}\,\mathrm{LS}_{n,1}(112) = [m]_q \prod_{i=2}^{m} (1+q^i)$$

is symmetric and unimodal of degree $d = (m^2+3m-4)/2$.

Proof. The displayed product formula is Dorward's Theorem 3.3, proved in Proposition 3.5 [1]; we use it as an attributed input. If $m = 1$, the product is 1 and the assertion is immediate. Assume $m \geq 2$.

Let W be the Weyl group of type B_m. Number its simple reflections $s_1, \ldots, s_m$ along the Dynkin diagram, with s_m at the short-root end, and set

$$J = \{s_1, \ldots, s_{m-2}, s_m\}.$$

Deleting s_{m-1} separates the diagram into types A_{m-2} and A_1. Here A_0 is the trivial group when $m = 2$. Consequently the standard parabolic subgroup W_J has type $A_{m-2} \times A_1$.

For a finite Weyl group U, write $F_U(q) = \sum_{w \in U} q^{\ell(w)}$, where ℓ is Coxeter length. The classical length products, with the empty-product convention, are

$$F_W(q) = \prod_{i=1}^{m} [2i]_q, \qquad F_{W_J}(q) = [2]_q \prod_{i=1}^{m-1} [i]_q.$$

These are the type B and type A formulas in [2, §§3–5]. Each right coset has a unique element v of minimum length, and every $w \in W$ has a unique factorization $w = vu$, $u \in W_J$, with $\ell(w) = \ell(v) + \ell(u)$. Thus the minimum-length representatives W^J have generating polynomial

$$\begin{aligned} F_{W^J}(q) &= \frac{\prod_{i=1}^{m} [2i]_q}{[2]_q \prod_{i=1}^{m-1} [i]_q} \\ &= \frac{[m]_q}{[2]_q} \prod_{i=1}^{m} (1+q^i) = [m]_q \prod_{i=2}^{m} (1+q^i). \end{aligned}$$

The cancellation uses $[2i]_q = [i]_q(1+q^i)$ and $[2]_q = 1+q$. Stanley's Theorem 3.1 states that the length coefficients of W^J are symmetric and unimodal for every parabolic quotient of a finite Weyl group. This proves the assertion for the displayed product. Its degree is

$$(m-1) + \sum_{i=2}^{m} i = \frac{m^2+3m-4}{2}.$$

Multiplication by q prepends a zero coefficient, which preserves unimodality since all coefficients are nonnegative. □

Why the classical theorem applies. The quotient above corresponds to the smooth complex projective homogeneous space $\mathrm{SO}(2m+1,\mathbb{C})/P$, where the Levi Weyl group of P is W_J. Its Schubert cells give $F_{W^J}(q) = \sum_{r=0}^{d} \dim H^{2r}(G/P,\mathbb{C})q^r$. Hard Lefschetz makes multiplication by an ample class injective from H^{2r} to H^{2r+2} for $r < d/2$; Poincaré duality gives symmetry. This is Stanley's argument, not a new proof of hard Lefschetz.

Scope and verification. For $n < 3$ there are no such restricted growth functions, so the polynomial is zero. We prove the coefficient assertion in Conjecture 4.1. Equality of rank polynomials does not identify Dorward's poset $M^1(m)$ with W^J, and no Sperner or symmetric-chain result for $M^1(m)$ is claimed. The accompanying exact verifier independently enumerates restricted growth functions of lengths 0 through 9 and minimum-length signed coset representatives for $2 \le m \le 10$. Those finite checks support the identification; the all-parameter proof is the argument above.

Tasks 2–5: proposed follow-ups

These are proposals, not additional solutions or campaign credits.

1. **Task 2.** Decide the Sperner assertion for Dorward's poset $M^1(m)$, which is not determined by its rank polynomial.

2. **Task 3.** Construct an explicit statistic-preserving bijection between the restricted growth functions and the signed coset representatives, with the one-unit shift in degree included.

3. **Task 4.** Construct symmetric chain decompositions of $M^1(m)$, or identify an obstruction to their existence.

4. **Task 5.** Determine which exact-k occurrence generating polynomials admit homogeneous-space interpretations or other Lefschetz operators, starting with $k = 2$.

References

[1] R. Dorward, *Combinatorial statistics on restricted growth functions containing a pattern exactly k times*, undergraduate thesis (internally published 2017), arXiv:2003.05073v1 (2020), Theorem 3.3 and Conjecture 4.1. https://arxiv.org/abs/2003.05073.

[2] R. P. Stanley, *Weyl groups, the hard Lefschetz theorem, and the Sperner property*, SIAM J. Algebraic Discrete Methods **1** (1980), 168–184, Theorem 3.1 and §§3–5. https://math.mit.edu/~rstan/pubs/pubfiles/42.pdf.